

PARTH PATEL

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EDUCATION

South Dakota State University

Bachelor of Science in Computer Science

• GPA: 3.5

Graduated: Dec 2025

Brookings, SD

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, JavaScript, Java, SQL
- **Frameworks & Libraries:** Flutter, React, Node.js, OpenCV, Arduino IDE
- **Development Tools:** Git, Docker, VS Code, Figma, Linux
- **Technologies:** REST API, IoT/Embedded Systems, Computer Vision, Machine Learning, BLE/USB Protocols
- **Cloud & DevOps:** AWS, HPC, Agile/Scrum methodologies

PROFESSIONAL EXPERIENCE

Integrated Sensing Circuits and Systems Laboratory, SDSU

Aug 2025 - Dec 2025

Undergraduate Research Assistant

- **Edema Monitoring System:** Developed an end-to-end data ecosystem for a custom lab-developed medical device, worked on embedded firmware, a Java-based Android app, and a Python PC GUI for real-time patient data visualization and saving in csv.
- **AED Version 1 & 2 Development:** Managed hardware-level assembly and hand-soldering of embedded units. Engineered cross-platform interfaces to capture, save, and graph complex electrochemical data, including Potentiometry, Impedance & more tests.
- **Firmware Engineering:** Programmed STM32 and Arduino microcontrollers to perform on-device calculations and data processing. Also, Integrated multi-protocol data transmission, utilizing Nordic BLE for mobile synchronization and UART for PC-based graphical interfaces.

Intelligent Vision Systems Laboratory, SDSU

May 2025 - Aug 2025

Undergraduate Research Assistant

- **Autonomous Spatial Mapping:** Developed a GPS-independent navigation system using Intel RealSense depth sensors and ArUco markers; implemented SLAM algorithms to calculate field boundaries and distances for automated equipment.
- **Document AI & Extraction:** Automated the conversion of research paper tables into structured text, trained models on diverse datasets to extract and summarize key parameters, significantly reducing manual data entry.
- **Model Optimization:** Streamlined image-to-text pipelines to improve the accuracy of tabular data recognition, facilitating faster research synthesis with lab members.

Plant Pathology Department, SDSU

May 2024 - Aug 2024

Undergraduate Research Assistant

- **Flutter Development & Automation:** Developed a custom Flutter mobile application to replace a time-intensive manual field entry process integrated features for high-resolution image uploads and automated CSV generation, streamlining data pipelines for leaf rust analysis.
- **Pathogen & Resistance Research:** Collaborated on studies to identify leaf rust resistance in barley varieties, managed the cultivation of bacterial cultures, and performed controlled inoculations to observe and record infection levels.
- **Greenhouse Management:** Facilitated the breeding of diverse barley species within controlled greenhouse environments, overseeing seed growth and sample preparation for further genetic and pathological analysis.
- **Digital Presence & Outreach:** Developed and launched the laboratory's website to showcase research findings, faculty profiles, and project milestones, significantly increasing the lab's visibility and professional presence.

LEADERSHIP & ACTIVITIES

Indian Students' Association, SDSU

Oct 2023 - Dec 2025

Treasurer

- Managed organizational budget and financial operations for 200+ member student organization
- Led fundraising initiatives and coordinated India Night cultural event, securing sponsorships, managing ticket sales, and organizing performances and catering for 300+ attendees

PROJECTS & ACHIEVEMENTS

No-Code REST API Builder (Senior Design Project): Designed drag-and-drop interface for creating REST APIs without manual coding, improving API development easily and Implemented visual workflow enabling non-technical users to design, test, and deploy functional APIs

Mobile Application Development:Built a mobile app to track work hours and total earnings across multiple jobs, featuring custom filters and automated pay reports.

CERTIFICATIONS & TRAINING

- Mastercard Cybersecurity Simulation: Forage (2023)